Nolan Aguirre

NolanAguirre08@gmail.com Github: NolanAguirre

Introduction

Full-stack systems engineer from Austin, TX.
Specializes in modular, composable architectures with deep experience across backend, infrastructure, and frontend.
Proven track record of building secure, high-scale, multi-tenant platforms from the ground up.
Thrive in early-stage environments by breaking ambiguity into core problems and delivering flexible, evolving systems

Languages & Tools: JavaScript, SQL, Shell, TypeScript, Python, Solidity, OCaml, HTML, CSS
Frameworks & APIs: React, GraphQL, REST, WebSockets, UDP
Databases: PostgreSQL, SQLite, MongoDB, memcached
Infra & DevOps: AWS, Linux, Git, Heroku, DNS, SSH, Unix user permissions, Terraform, Docker
Security: OAuth2, JWT, field-level encryption, RBAC, SSH
Blockchain: Ethereum, Tezos, Aptos

Experience

VP of Engineering, AutoHDR *October 2025 — January 2026*

AI Real-Estate photo editing for home listings.

- Advise founders in a large range of topics, from product, security, hiring, communication, and more.
- Introduce kanban board for project management, team all hands meetings, one on ones, and more.
- Develop full stack monorepo to accelerate feature development. Monorepo used Terraform with AWS, Heroku, Javascript, React, postgresql, nginx with Makerules and shell scripts for automation.
- Begin refactor around microservices architecture and improved security.
- Lead hiring resulting in two developers.

Co-founder, JNLabs *May 2024 — May 2025*

Formed to identify and test early-stage product opportunities in the AI space.

- Collaborate closely with co-founder to identify and strategize AI-driven market opportunities, focusing on product discovery and user engagement.
- Rapidly prototype and launch AI projects to assess real-world user interest, enabling market-driven insights and informed product decisions.
- Developed AIPOP, a Nodejs based AI music generation app for the Taiwanese market, integrated within LINE, providing unique, accessible music experiences.
- Oversee and execute essential tech operations for the company, administrating various buisnesss accounts such as AWS, Gmail, etc.

Founding engineer, Vertalo *April 2019 — July 2025*

Fintech platform for private asset management; 100k+ investors in SEC-regulated private securities.

- Led architecture and implementation of a secure, multi-tenant financial platform spanning backend, frontend, infrastructure, and blockchain integration
- Built a modular system around a double-entry ledger with full audit tracking (who, when, what), powering all state transitions and financial events
- Designed a general-purpose event queue with ordered, replayable, write-once semantics; paired with an event reconciliation system to safely convert external triggers (e.g., account claim tokens, blockchain actions) into in-platform effects
- Developed role-based APIs for investors, asset managers, administrators, ATS platforms, and agents; enabled inter-role sharing and RBAC tiering (viewer/admin/owner) across all surfaces
- Built a graph-based onboarding engine with reusable qualifications, OR-logic prerequisites, single-use gating, and per-manager schema-driven form customization
- Integrated with third-party platforms using OAuth and client-provided API keys, including DocuSign, North Capital, Coinbase, Netki (KYC) and Investready(AML)
- Deployed smart contracts in Solidity (Ethereum) and OCaml (Tezos); built a 500M+ record Ethereum block indexer and versioned parser system for contract data
- Implemented field-level encryption, soft-delete policies, full historical audit logging, a built-in dead letter queue with human alerting, and automated schema migration via Sqitch
- Built internal tooling for logging, notification dispatch, admin overrides, UI generation and an in house testing framework